SHARDUL JOSHI

B.ENG. MECHATRONICS

ABOUT ME

I am a motivated mechatronics student with a strong foundation in programming, data analysis, and automation. With hands-on experience in embedded systems and practical project work.

CONTACT

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TECHNICAL SKILLS

- MATLAB & Simulink
- C++
- Python
- CAD
- ROS
- Microsoft Office
- Node-RED
- Computer Vision
- Embedded Systems
- Machine Learning
- Git

LANGUAGES

- English – Fluent
- German – Working Proficiency
- Hindi – Fluent

EDUCATION

OCTOBER 2025 – PRESENT
MASTER OF SCIENCE IN MECHATRONICS AND ROBOTICS(M.SC.),
LEIBNIZ UNIVERSITÄT HANNOVER

OCTOBER 2021 – AUGUST 2025
BACHELOR OF ENGINEERING IN MECHATRONICS (B.ENG.),
THWS – TECHNISCHE HOCHSHULE WÜRZBURG-SCHWEINFURT

CURRENTLY IN 7TH SEMESTER.
Specializations:

- Robotics and Production
- Applied Machine Learning and Design of Experiments
- Automated Systems and Human-Machine Interaction

EXPERIENCE

1ST JUNE 2024 – 30TH NOVEMBER 2024
INTERN,
ROBERT BOSCH GMBH

- Focused on **automating** and **interfacing** multiple devices within a testbench used to **validate** Electronic Control Units **(ECUs)**
- Established **seamless communication** and **device coordination** using diverse **communication protocols**, addressing the challenge of synchronizing multi-device operation
- Developed **automation scripts** and a Python-based **GUI** using libraries such as **Tkinter**, enabling **real-time control** and monitoring of devices on the testbench
- Achieved efficient **testbench automation**, improving workflow by **reducing** manual intervention and ensuring consistent, accurate device behaviour and **data collection**.

1ST MARCH 2023 – 15TH FEBRUARY 2024
PROGRAMMING TUTOR,
THWS – TECHNISCHE HOCHSHULE WÜRZBURG-SCHWEINFURT

- Provided tutoring support for the **Programming 2** course, focusing on **object-oriented programming** (OOP) concepts in **C++.**
- **Helped** students understand complex OOP principles, addressing difficulties with topics like **inheritance**, **polymorphism**, and **encapsulation**.

PERSONAL SKILLS

- Public Speaking
- Analytical thinking
- Hardworking

INTERESTS

- Hiking
- Bouldering
- Volleyball
- Travelling

PROJECTS

BACHELOR THESIS,

DEVELOPMENT OF AN POWER MONITORING SYSTEM FOR INDUSTRIAL USECASE

- Developed power monitoring system which transmits consumption data **wirelessly** in a **WPA enterprise** network.
- **Two distinct prototypes** were developed and tested across multiple metrics such as **accuracy, cost, data resolution**, etc.
- Accuracy of about **4%** was achieved in one method and the other method had an accuracy of **0.5%.**
- Thesis was concluded as a **comparative study** with deep insights into the **advantages and shortcomings** of each developed approach.

MINI-PLC,

THWS – TECHNISCHE HOCHSHULE WÜRZBURG-SCHWEINFURT

- Developed a mini-PLC system using **Arduino**, integrating **power electronics** to replicate the **functionality** of a real Programmable Logic Controller (**PLC**).
- Faced the challenge of **designing** robust power electronics and ensuring reliable communication and **control**, similar to an industrial PLC.
- **Designed** and manufactured **custom circuits**, created a **PCB** using **Altium Designer**, and fabricated a **3D-printed** enclosure using **Fusion 360** to house the system.
- Successfully built a **fully functional mini-PLC**, achieving a compact and reliable system with a professional-grade PCB and custom 3D-printed housing.